

DynamOS: The Operating System for AI-Native Education

A Platform for Creator-Educators

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Date: May 2026

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1. Executive Summary

In an era where Artificial Intelligence can compress knowledge on demand, the purpose of education must evolve from information transfer to metacognitive mastery. We must teach students how to think structurally about learning itself. DynamOS is the platform built for this new paradigm. It is a cognitive operating system that inverts the current model: instead of AI teaching students, DynamOS empowers students and teachers to teach the AI, and in doing so, build their own powerful, personalized learning tools.

We call this model **"Roblox for Education."** Just as Roblox gave millions of children the tools to become game developers, DynamOS gives educators and learners a no-code "Studio" to build, share, and even monetize AI-powered educational mini-apps. These are not static worksheets, but dynamic, interactive experiences—from Socratic tutors to adaptive quizzes and spaced repetition systems—all grounded in established pedagogical science.

The core of DynamOS is the RUA (Recognize, Understand, Apply) framework, a metacognitive engine that structures every interaction. This pedagogical foundation is our primary moat, distinguishing DynamOS from generic AI wrappers. We are not building a better textbook; we are building the printing press.

Our business model is a hybrid of Freemium SaaS for creators and a revenue-sharing marketplace for their creations, targeting a global market of over 85 million educators. This document outlines our vision, architecture, and strategy to build the definitive platform for the next generation of teaching and learning.

2. The Problem: A Crisis of Tools and Agency

The education sector is facing an inflection point, caught between outdated industrial-era models and a tidal wave of unguided AI. This tension creates several critical problems that DynamOS is designed to solve:

- **Tools Built by Engineers, Not Educators:** The majority of ed-tech tools are designed by software engineers with little to no classroom experience. The result is a landscape of rigid, expensive, and pedagogically shallow products that fail to meet the dynamic needs of real teachers and students.
- **AI without a Framework:** Educators are experimenting with tools like ChatGPT in isolation, creating a chaotic and inefficient "prompt-engineering wild west." Without a shared pedagogical framework, the quality of AI integration is inconsistent, and AI is often perceived as a threat to be managed rather than a tool to be wielded.
- **The "App Gap":** An educator who needs a specific tool—a customized vocabulary drill for their curriculum, for instance—has no viable path. Learning to code is impractical, and custom development is prohibitively expensive. This stifles innovation at the classroom level.
- **One-Size-Fits-None Learning:** Traditional and digital curricula are delivered uniformly, failing to account for the diverse needs of individual learners. Personalization remains a buzzword, not a reality, because scalable, teacher-driven customization is impossible with current tools.

In short, the very people on the front lines of education lack agency over their digital tools. DynamOS restores this agency, transforming educators from passive consumers of technology into active creators of learning experiences.

3. The Vision: Roblox for Education

Our vision is to build the platform where the world's teachers become the world's most effective ed-tech developers—without writing a single line of code. The most powerful analogy for this is Roblox.

Roblox	DynamOS
Roblox Studio (game builder)	DynamOS Studio (prompt + app builder)
Lua scripting	RUA prompt engineering framework
Millions of user-built games	Millions of teacher-built learning apps
Robux economy	DynamOS Credits + revenue sharing

Roblox succeeded not by making the best games, but by making game creation accessible to everyone. DynamOS applies this same architecture to learning itself. We believe that the act of designing a learning experience—of prompt engineering a Socratic tutor or building an adaptive quiz—is a profound metacognitive exercise. We are creating a platform where the act of creation is, in itself, an act of learning.

Our North Star: By 2030, DynamOS powers 10% of all digital learning interactions globally, built not by corporations, but by the teachers who know their students best.

This is a fundamental shift from content consumption to active creation. It is a move from static knowledge to dynamic systems of learning. This is the future of AI-native education.

4. Product Architecture: The Five Layers of DynamOS

DynamOS is not a single app, but a layered platform where each component builds upon the last, creating a comprehensive ecosystem for AI-native education. It is designed to be coherent, scalable, and pedagogically sound.

Layer 1: AI-Native Content Engine

This is the core pedagogical processor of the system. It is not a generic LLM wrapper. Every interaction is filtered through a set of pedagogical models, with Kee Lee's RUA Framework (Recognize → Understand → Apply) as the master algorithm. This engine ensures that content generation, feedback, and scaffolding are grounded in learning science, referencing methodologies like Bloom's Taxonomy, Task-Based Language Teaching (TBLT), Communicative Language Teaching (CLT), and the Lexical Approach.

Layer 2: Creator Studio

This is the "Roblox Studio" for educators. A no-code, visual app builder allows teachers to assemble "DynApps" from pre-built components (e.g., quiz, flashcard, tutor, simulation). At its heart is a visual prompt engineering framework that makes the logic of AI interaction transparent and malleable. Teachers are not just writing prompts; they are designing learning workflows and feedback loops.

Layer 3: Student Experience

For students, DynamOS is an adaptive, engaging learning environment. It delivers personalized learning paths, providing scaffolding within their Zone of Proximal Development (ZPD). Core features like spaced repetition for memory consolidation and AI-

driven feedback for skill application are built into every DynApp, ensuring an experience that is both effective and motivating.

Layer 4: Teacher Dashboard

This is the mission control for educators. The dashboard provides deep, real-time analytics on student progress, moving beyond simple metrics like "percent correct." It offers formative assessment insights, highlighting where students are struggling (e.g., at the 'Understand' stage of RUA) and suggesting interventions. It makes the invisible process of learning visible.

Layer 5: Marketplace

The marketplace is the engine of our ecosystem's growth. Here, creators can publish, share, and monetize their DynApps. Other educators can discover, rate, and—critically—remix existing apps for their own classrooms. This fosters a collaborative culture of continuous improvement and allows the best pedagogical approaches to scale organically.

5. Pedagogical Foundation: An OS Built on Learning Science

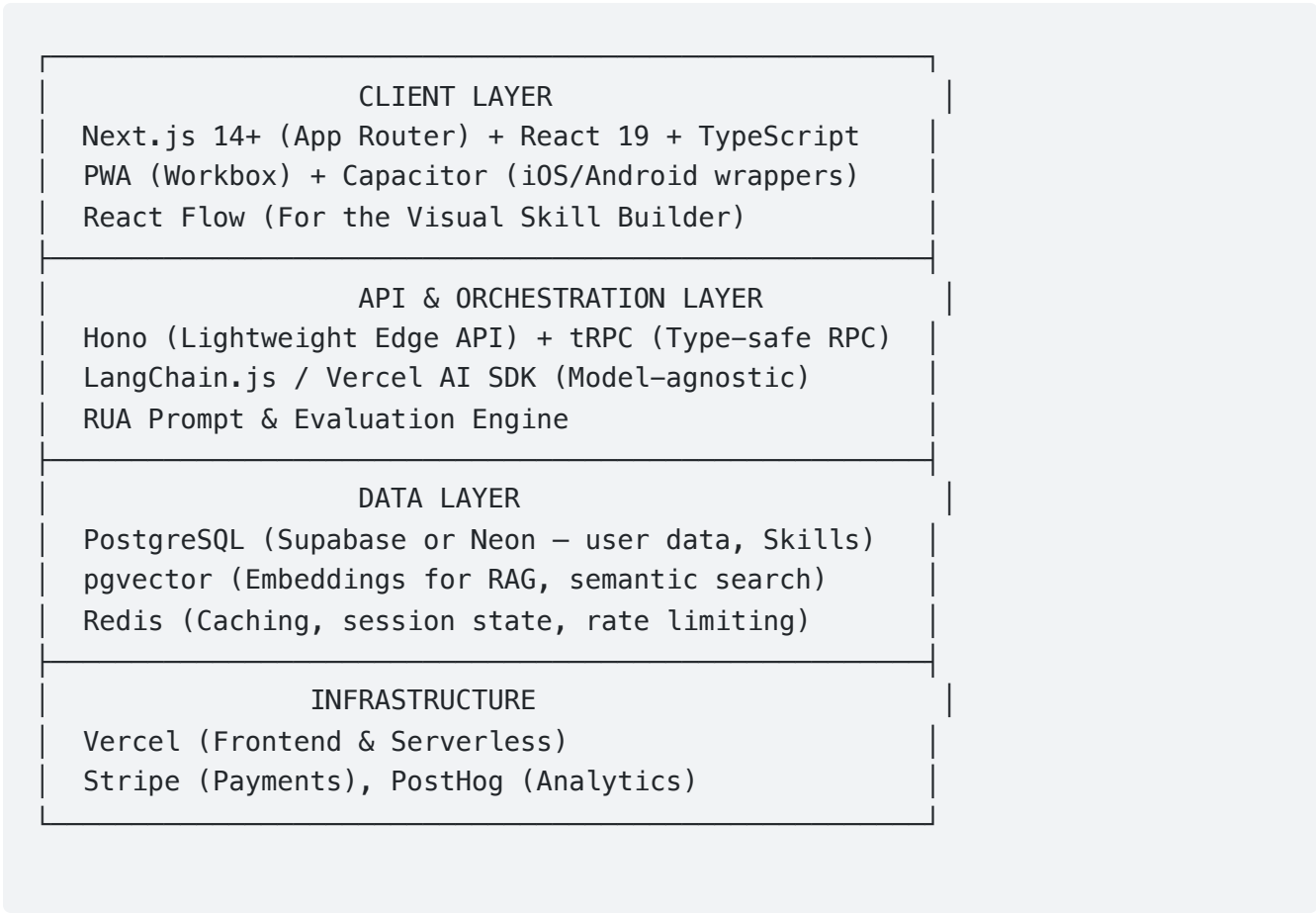
DynamOS is not "tech-first"; it is "pedagogy-first." Our architecture is a direct translation of established learning science into a digital framework. AI is the enabling technology, but the following principles are the true foundation of the platform.

- **RUA Framework (Recognize → Understand → Apply):** Kee Lee's core methodology, which posits that all durable learning moves through these three stages. This framework structures all content, assessment, and feedback within DynamOS.
- **Bloom's Taxonomy:** The Creator Studio guides educators to design activities that target specific cognitive levels, from Remembering and Understanding to Analyzing, Evaluating, and Creating.
- **Vygotsky's Zone of Proximal Development (ZPD):** The AI acts as a "More Knowledgeable Other," providing automated scaffolding to help learners bridge the gap between what they can do alone and what they can achieve with guidance.
- **Krashen's i+1 Input Hypothesis:** In language learning, the platform continuously assesses a student's level (i) and delivers content and tasks that are just one step beyond (i+1), ensuring optimal challenge.
- **Ebbinghaus Forgetting Curve / Spaced Repetition:** The system intelligently schedules review sessions for memorization-based tasks, combating the natural decay of memory and ensuring long-term retention.
- **Task-Based & Communicative Language Teaching (Willis, Hymes):** For language apps, the focus is on completing meaningful tasks and achieving communicative competence, rather than rote memorization of grammar rules.
- **Lexical Approach (Lewis):** The platform emphasizes the learning of language in chunks (collocations, fixed expressions) as they naturally occur, rather than as isolated words.

- **TPACK (Technological Pedagogical Content Knowledge):** DynamOS is a tool designed to fuse these three knowledge areas, empowering teachers to use technology in a way that is deeply integrated with their pedagogical skills and subject matter expertise.
- **Self-Regulated Learning & Metacognition:** By making the structure of learning visible in the Creator Studio, DynamOS encourages both teachers and students to think about **how** they learn.
- **Mastery Learning (Bloom):** The platform's adaptive nature allows for a mastery-based progression, where students must demonstrate competence on a concept before moving to the next, ensuring no foundational gaps are left behind.

6. Technical Architecture: Built for Trust and Scale

The DynamOS technical architecture is designed to be robust, scalable, and model-agnostic, with a core emphasis on data privacy and user trust.



Core Principles

- **Model-Agnostic:** The orchestration layer is designed to work with any major AI provider (OpenAI, Anthropic, DeepSeek, Gemini, etc.) as well as open-source local models. This

prevents vendor lock-in and allows creators to choose the best model for their specific task.

- **Privacy by Design (BYOK):** A key feature is the "Bring Your Own Key" (BYOK) model. This allows schools and individuals to use their own AI provider accounts. In this mode, DynamOS never sees or stores the content of prompts and responses, which provides a critical layer of trust and simplifies compliance with regulations like FERPA, GDPR, and COPPA.
- **Scalable Serverless Infrastructure:** By leveraging serverless functions (Vercel) and managed databases (Supabase/Neon), the platform can scale efficiently from a single classroom to millions of users without significant upfront infrastructure investment.
- **Data-Driven Pedagogy:** While respecting privacy, the platform aggregates anonymized interaction data to identify which pedagogical patterns are most effective. This allows us to continuously improve the Creator Studio's templates and recommendations, creating a powerful data flywheel.

7. Market Analysis: A Watershed Moment

The education market is at a unique inflection point where four major trends are converging, creating an unprecedented opportunity for a platform like DynamOS.

- 1. **AI is Ready:** Models of the GPT-4 class are now powerful enough to handle nuanced educational tasks, and their costs are rapidly decreasing.
- 2. **Teachers are Experimenting:** Educators globally are already using AI, but need structured, pedagogically-sound tools to move beyond simple chat.
- 3. **The Creator Economy is Mature:** Platforms like Substack and Roblox have proven the "build once, monetize forever" model. This has not yet been applied effectively to education.
- 4. **RUA is the Missing Link:** The world has AI tools, but lacks a pedagogical framework to make them work specifically for learning. RUA provides this crucial layer.

Market Size

Segment	Size	Growth
Global K-12 & Higher-Ed Educators	85M+	~2.5% YoY
AI in Education Market	\$4.0B (2025) → \$30B+ (2030)	36% CAGR
Initial Addressable Market (English-speaking educators)	15M	10 -

The initial go-to-market will also focus on the highly motivated private tutoring market in Asia, particularly within the China training school ecosystem, where demand for innovative, effective learning tools remains exceptionally high post-regulatory changes.

8. Business Model: A Hybrid SaaS & Marketplace

Our revenue model is designed to maximize adoption through a generous free tier while capturing value from professional users and the marketplace ecosystem. It consists of two primary streams: SaaS subscriptions and a marketplace commission.

SaaS Tiers

- **Free Tier:** Allows educators to build a limited number of DynApps and use them with their students. This is the top of our funnel and the engine of our organic growth. All users must bring their own API key (BYOK).
- **Teacher Pro (\$15/month):** Aimed at power users and creators, this tier offers access to the full Creator Studio, unlimited app creation, advanced analytics, and priority access to new features.
- **School/Institutional License:** Custom pricing for schools and districts, offering SSO, admin dashboards, LMS integration, and centralized management.

Marketplace Revenue Share

The DynamOS Marketplace allows creators to monetize their premium DynApps and prompt packs. We believe in empowering our creators, and our revenue structure reflects this:

- **Creator Payout:** 70% of the revenue from a sale goes directly to the creator.
- **Platform Commission:** 30% is retained by DynamOS to cover platform costs, moderation, and transaction fees.

This hybrid model creates a powerful flywheel: the free SaaS tier attracts a large base of users, a portion of whom become Pro subscribers and creators. These creators populate

the marketplace with high-quality content, which in turn attracts more users and provides a scalable, high-margin revenue stream.

9. Go-to-Market Strategy:

Seeding a Creator Ecosystem

Our go-to-market strategy is focused on winning the trust and enthusiasm of innovative educators first, knowing that they will become the evangelists who drive mainstream adoption.

Phase 1: Seed the Creator Community (Months 1-6)

We will manually recruit an initial cohort of 500 "founding creator-teachers" from ed-tech influencer circles and alumni of Kee Lee's RUA workshops. They will receive free lifetime Pro accounts and enhanced revenue-sharing terms in exchange for populating the marketplace with the first wave of high-quality DynApps.

Phase 2: Organic Growth Flywheel (Months 7-18)

We will leverage the natural growth loop of the platform: teachers create apps and share them with students, who in turn expose the platform to other teachers and parents. This product-led growth will be supplemented by a "DynamOS Ambassador" program, content marketing focused on AI pedagogy, and a strong presence at key education conferences like ISTE and BETT.

Phase 3: Institutional Sales (Months 19-36)

With a thriving marketplace and proven efficacy, we will begin a targeted sales motion into schools and districts. Case studies from our early adopters will be crucial in demonstrating the platform's value and driving larger institutional contracts. A parallel strategy will target the Chinese market, carefully positioned to align with local regulations by focusing on "AI literacy" and self-directed learning tools rather than prohibited tutoring.

10. Competitive Moat: The RUA Engine & Network Effects

While many companies can wrap a UI around an AI model, DynamOS's defensibility comes from two compounding moats that are difficult to replicate.

Moat 1: The Pedagogical Framework

Our most significant moat is the RUA engine and the deep integration of learning science. Competitors can copy features, but they cannot easily copy a pedagogical philosophy that is embedded in the platform's architecture and the community's DNA. Every tool, template, and analytic in DynamOS is designed to reinforce the Recognize-Understand-Apply loop. This creates a cohesive, effective user experience that generic platforms cannot match and establishes a brand trusted by serious educators.

Moat 2: The User-Generated Content & Remix Network

Our second moat is the network effect of the marketplace, modeled on platforms like Roblox and Figma.

- **Content Network Effect:** The more high-quality apps created, the more valuable the platform becomes for consumers.
- **Creator Network Effect:** The more consumers on the platform, the more attractive it is for creators.
- **The "Remix" Flywheel:** Allowing users to fork and improve upon existing apps dramatically accelerates innovation. A great lesson plan can be adapted for a dozen different contexts in a matter of hours, creating a content library that is both vast and hyper-relevant. This is a powerful advantage that closed, top-down content platforms like Khan Academy cannot replicate.

This combination of a unique pedagogical core and a powerful, compounding network effect makes our position highly defensible against both incumbent ed-tech players and large, horizontal AI companies.

11. Development Roadmap

Our development is phased to prioritize validating the core concept first, then building the community, and finally scaling the platform.

Phase 1: MVP "CoreOS" (Months 1-6)

The goal is to build the core creation and consumption loop. This includes the Prompt Studio with the RUA template engine, the first four app types (Quiz, Flashcard, AI Tutor, Interactive Story), a basic Classroom OS for assigning and tracking, and a marketplace for free apps. Monetization is not part of the MVP.

Phase 2: Public Launch & Community Growth (Months 7-12)

This phase focuses on launching monetization and community features. We will introduce the Teacher Pro subscription and enable paid apps in the marketplace. We will add more app types, release a mobile PWA, and integrate with Google Classroom. The RUA analytics dashboard will be a key feature release.

Phase 3: Platform Scale & Enterprise (Year 2)

The focus shifts to scaling and serving institutions. We will build out the enterprise/school tier with SSO and admin features, introduce a white-label option for publishers, and support multiple languages, starting with Chinese and Spanish. The goal is to establish DynamOS as critical infrastructure for AI-native education.

12. Team: Pedagogy-First

The founding vision of DynamOS is driven by an educator, not just an entrepreneur. This grounds our product development in the authentic needs of the classroom.

Kee Lee, Founder & Chief Learning Officer

Kee Lee brings over a decade of experience in teaching and curriculum design, particularly in the demanding fields of language acquisition and test preparation. He is the creator of the RUA framework, which was developed and refined through thousands of hours of real-world teaching. His deep understanding of pedagogical science and the practical challenges faced by educators is the guiding force behind the platform's architecture. He has a proven track record of creating effective and engaging learning systems, including the popular DynaSaurus AI Language Tutor and IELTS with Kee platforms, both of which serve as early prototypes and validation for the DynamOS concept.

We are currently in the process of recruiting a technical co-founder who shares our passion for education and has experience in building scalable, consumer-grade platforms with integrated AI/ML systems.

13. Appendix: Frameworks & Glossary

Glossary

- **DynamOS:** The educational operating system/platform.
- **DynaSaurus:** An example of a "DynApp" built on the DynamOS platform; an AI Language Tutor.
- **DynApp:** A mini-application created within the DynamOS Creator Studio.
- **RUA Framework:** Recognize, Understand, Apply. The core pedagogical model of DynamOS.
- **Creator Studio:** The no-code environment where educators build DynApps.
- **BYOK (Bring Your Own Key):** An operating mode where users connect their own AI provider accounts for enhanced privacy and control.

Pedagogical Frameworks Cited

- **Bloom's Taxonomy:** A hierarchical model for classifying educational learning objectives into levels of complexity and specificity.
- **Communicative Language Teaching (CLT):** An approach to language teaching that emphasizes interaction as both the means and the ultimate goal of study.
- **Lexical Approach:** A method of teaching foreign languages that prioritizes vocabulary and lexical chunks as the main focus of learning.
- **Task-Based Language Teaching (TBLT):** An approach centered on the completion of meaningful tasks, where language is a tool to be used, rather than an object to be studied.

- **TPACK (Technological Pedagogical Content Knowledge):** A framework that identifies the kinds of knowledge teachers need to teach effectively with technology.
- **Zone of Proximal Development (ZPD):** A concept by Lev Vygotsky referring to the difference between what a learner can do without help and what they can achieve with guidance.